WHEN STABLE EXPLANATIONS DISAGREE

A Stability–Localization Disagreement Audit for Chest X-ray Classification

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RESEARCH QUESTION & MOTIVATION

Can an explanation remain highly stable under image corruption while losing alignment with annotated opacity regions?

- AUDIENCE ML researchers, XAI practitioners, and medical-imaging AI developers.
- PURPOSE Test whether attribution rankings can remain stable despite losing alignment with radiologist-annotated opacity regions.
- SCOPE Exploratory audit of one ResNet-50, three XAI methods, and six blur/noise conditions. No clinical validation.

EXPERIMENTAL DESIGN

DATA & MODEL

RSNA chest X-rays; binary lung-opacity classification. ResNet-50 test AUROC = 0.879.

EVALUATION

300 held-out images (150 positive / 150 negative); Gaussian blur and noise at three severities each.

EXPLANATIONS

Integrated Gradients, Score-CAM, Occlusion Sensitivity. Original predicted class held fixed as the explanation target.

AUDIT DEFINITION

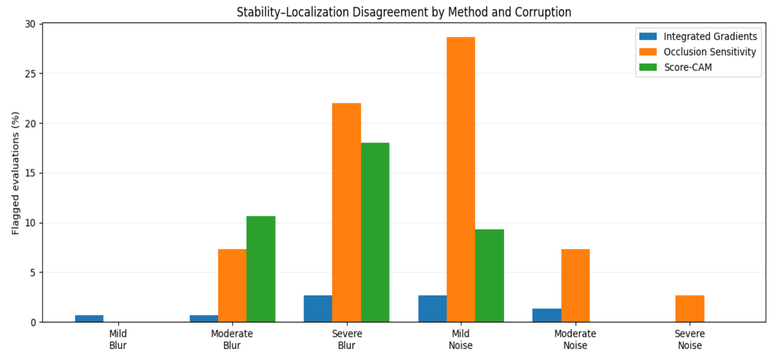
FLAG AN OBSERVATION WHEN BOTH HOLD:

① Stability ≥ 0.80 (Spearman similarity of attribution rankings)

② Localization change ≤ −0.05 (≥ 5 percentage-point decrease in attribution mass inside annotated boxes)

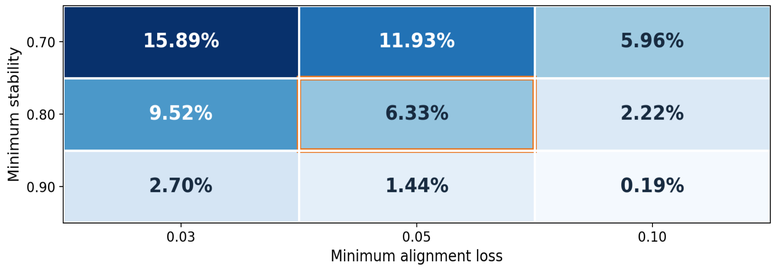
Audit applies only to 150 annotated positive images; 2,700 repeated method–image–corruption observations.

WHERE DISAGREEMENTS OCCUR



Peak: Occlusion Sensitivity + mild noise = 43/150 (28.67%).

THRESHOLD SENSITIVITY



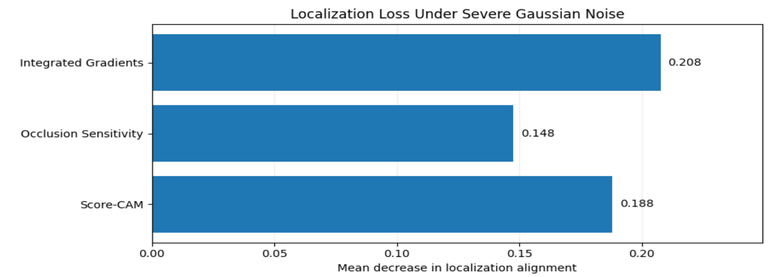
INTERPRETING THE AUDIT

171 / 2,700 positive-image evaluations (6.33%) flagged.

169 / 171 flagged evaluations (98.83%) retained the predicted class.
Among 1,076 high-stability evaluations, 171 (15.9%) also lost ≥5 percentage points of localization alignment.

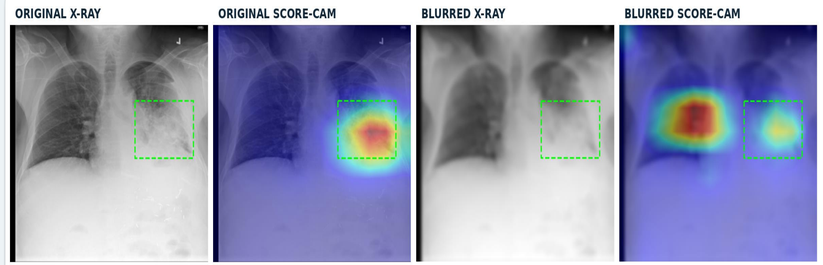
Counts represent method × image × corruption observations, not unique patients.

MEAN LOCALIZATION ALIGNMENT LOSS UNDER SEVERE NOISE



Loss occurred in 96.7% (IG), 88.7% (Occlusion), 94.0% (Score-CAM).

IMAGE CASE STUDY



Validated case: Score-CAM · severe blur (σ=3.0) · prediction unchanged
Stability 0.810 · alignment 0.488 → 0.161 (−0.327)

TAKEAWAY, LIMITATIONS & NEXT STEPS

MAIN FINDING: Stable attribution rankings can coexist with poorer opacity-box alignment and unchanged predictions.

IMPLICATION: Assess stability and localization jointly; stability alone can conceal deterioration.

LIMITATIONS: One dataset/model; bounding-box proxy; exploratory thresholds; no clinical validation.

CONTRIBUTION : Reproducible stability–localization disagreement audit with method-specific results and threshold sensitivity.

EVIDENCE : 150 annotated positive images × 3 XAI methods × 6 corruptions = 2,700 repeated evaluations • 171 flagged (6.33%) • 169/171 predictions unchanged

DATA / METHODS: RSNA Pneumonia Detection Challenge (2018) • ResNet-50 • Integrated Gradients • Occlusion Sensitivity • Score-CAM • Exploratory thresholds, not clinically calibrated.